

GWDD Model Verification Dashboard

U.S. Gas-Weighted Degree Day forecasts vs GFS ECE GEFS vs. observed history

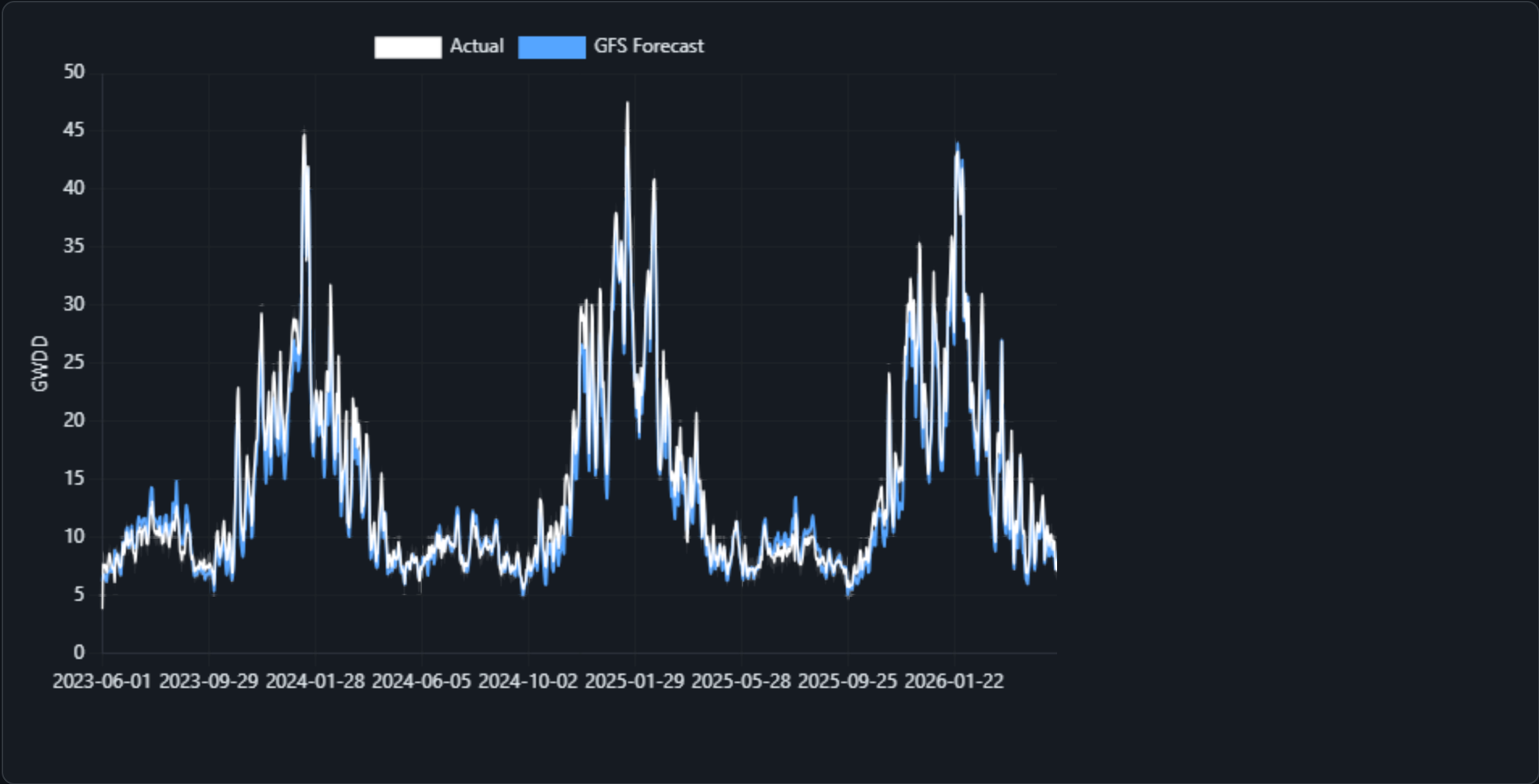
Model Scorecard

Summary stats per model &— lead-time window. Cells are tinted green (better than peers in the same window) to red (worse).

Model	Window	MAE	RMSE	Bias	Correlation	Â±1 Hit %	Â±2 Hit %	Mean Chg1	N
GFS	1-5d	1.184	1.521	-0.743	0.989	52.4	82.0	0.303	5330
ECE	1-5d	0.984	1.195	-0.408	0.994	55.9	93.4	0.166	5158
GEFS	1-5d	0.892	1.177	-0.546	0.992	65.1	91.7	0.235	5102
GFS	6-10d	1.953	2.826	-0.566	0.944	39.3	65.9	1.325	5327
ECE	6-10d	1.545	2.146	-0.260	0.970	43.9	75.2	â€”	5131
GEFS	6-10d	1.560	2.312	-0.377	0.964	49.0	75.3	0.502	5016
GFS	11-15d	2.824	4.203	-0.431	0.863	31.2	54.5	2.448	4791
ECE	11-15d	2.135	3.078	-0.159	0.931	37.4	64.4	â€”	3558
GEFS	11-15d	2.196	3.258	-0.223	0.923	38.8	63.6	0.642	4433

Forecast vs Actual

Time series of mean forecast within the selected lead-time window, overlaid on observed GWDD with a ± 1 GWDD band.



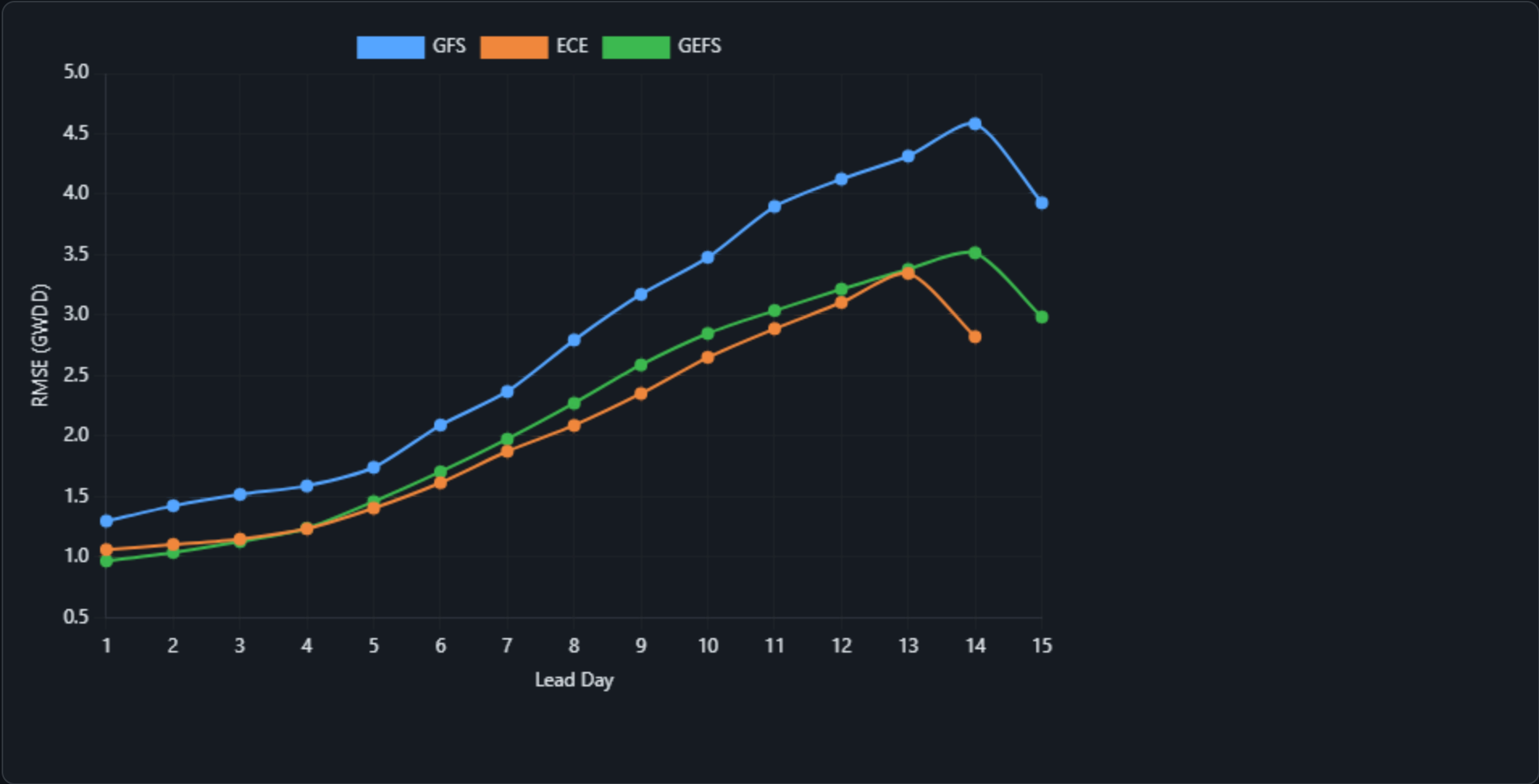
Bias & Seasonality

Mean (forecast $\hat{a}$ ~ actual). Positive = model forecasts higher GWDD than realized (cold bias on demand). Bars with $|\text{bias}| > 1$ GWDD are red-bordered.



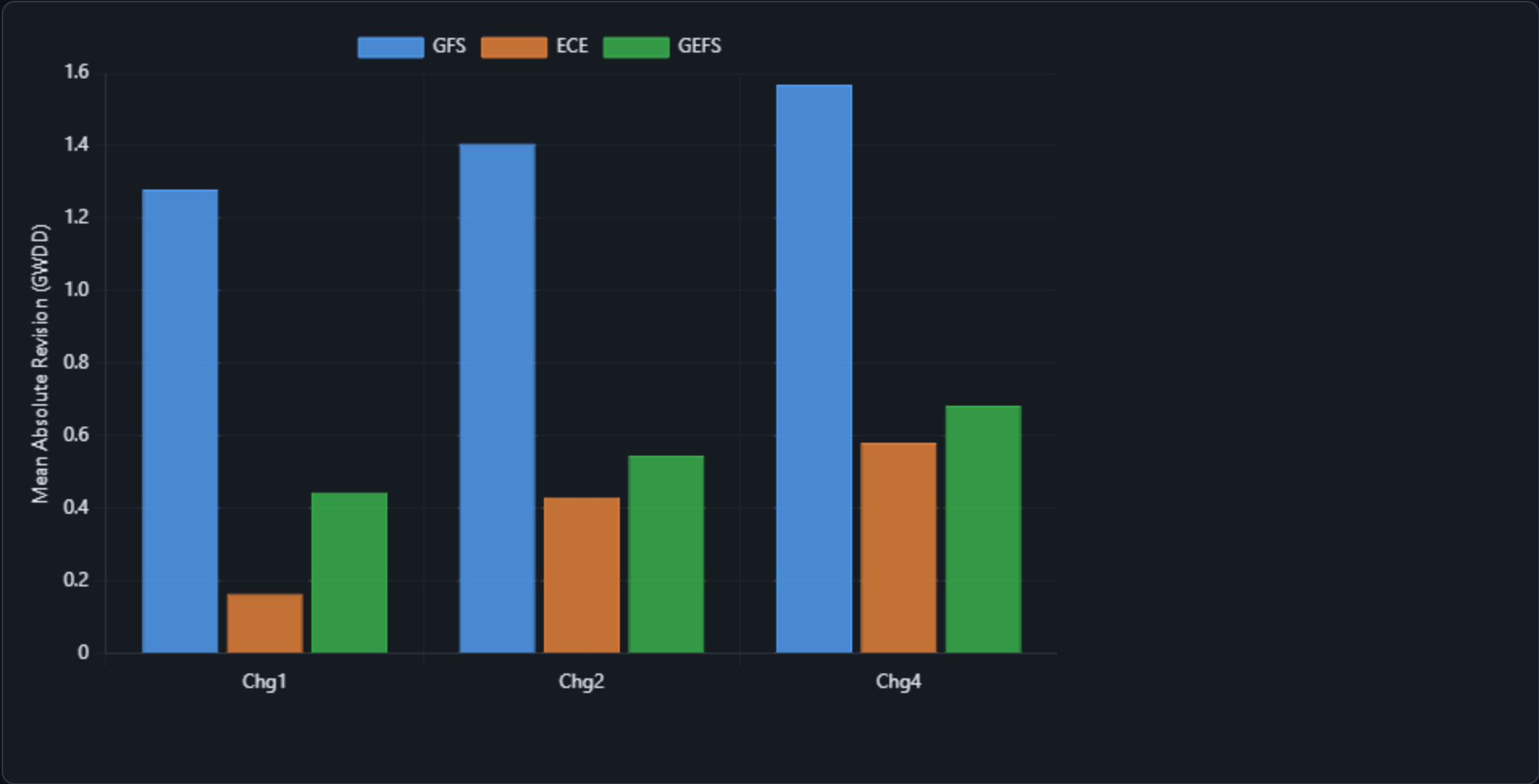
Model Convergence

Average RMSE by lead day. Error typically grows as the lead time extends.



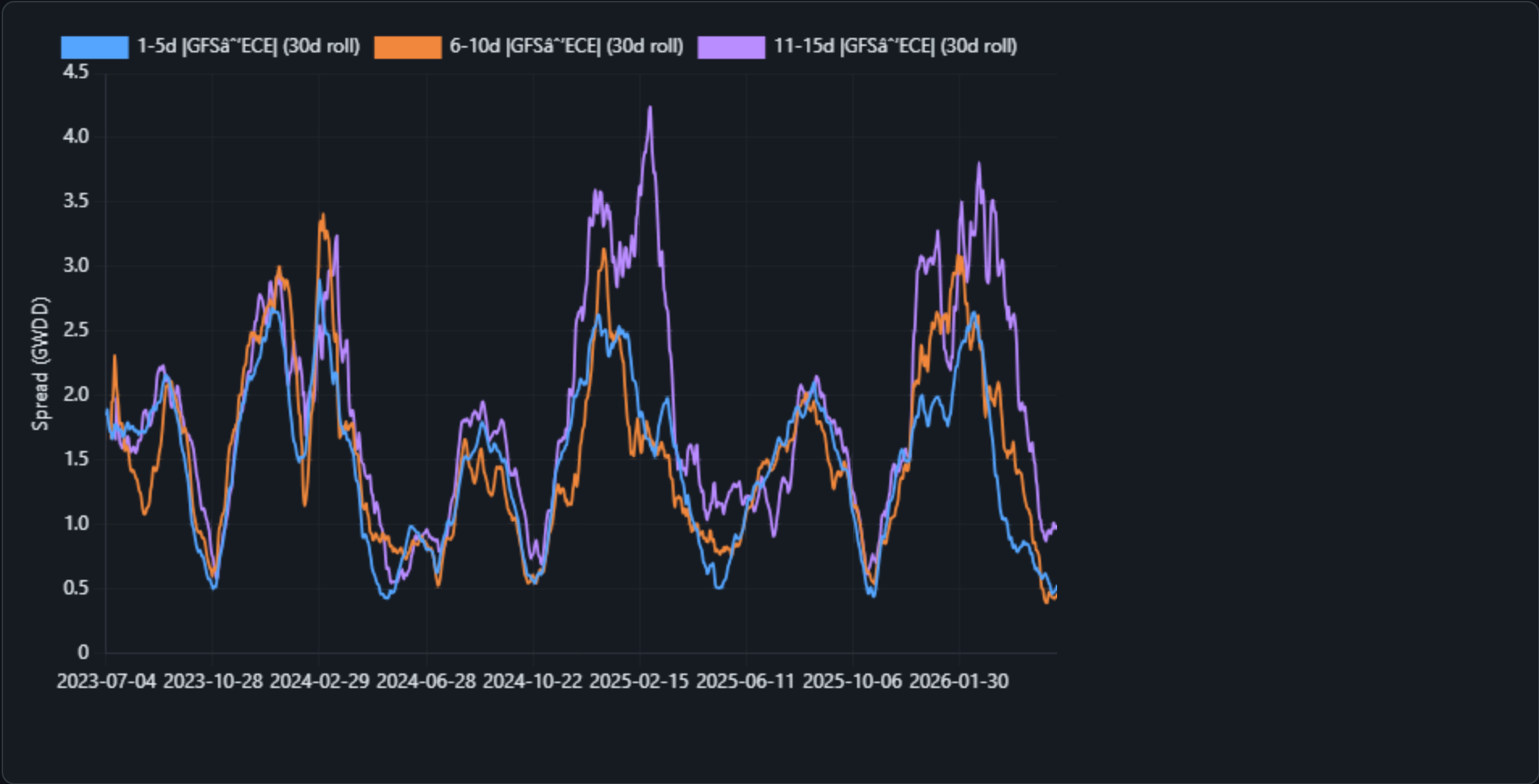
Forecast Revisions

Mean absolute run-over-run change. Higher = more volatile guidance; lower = more stable.



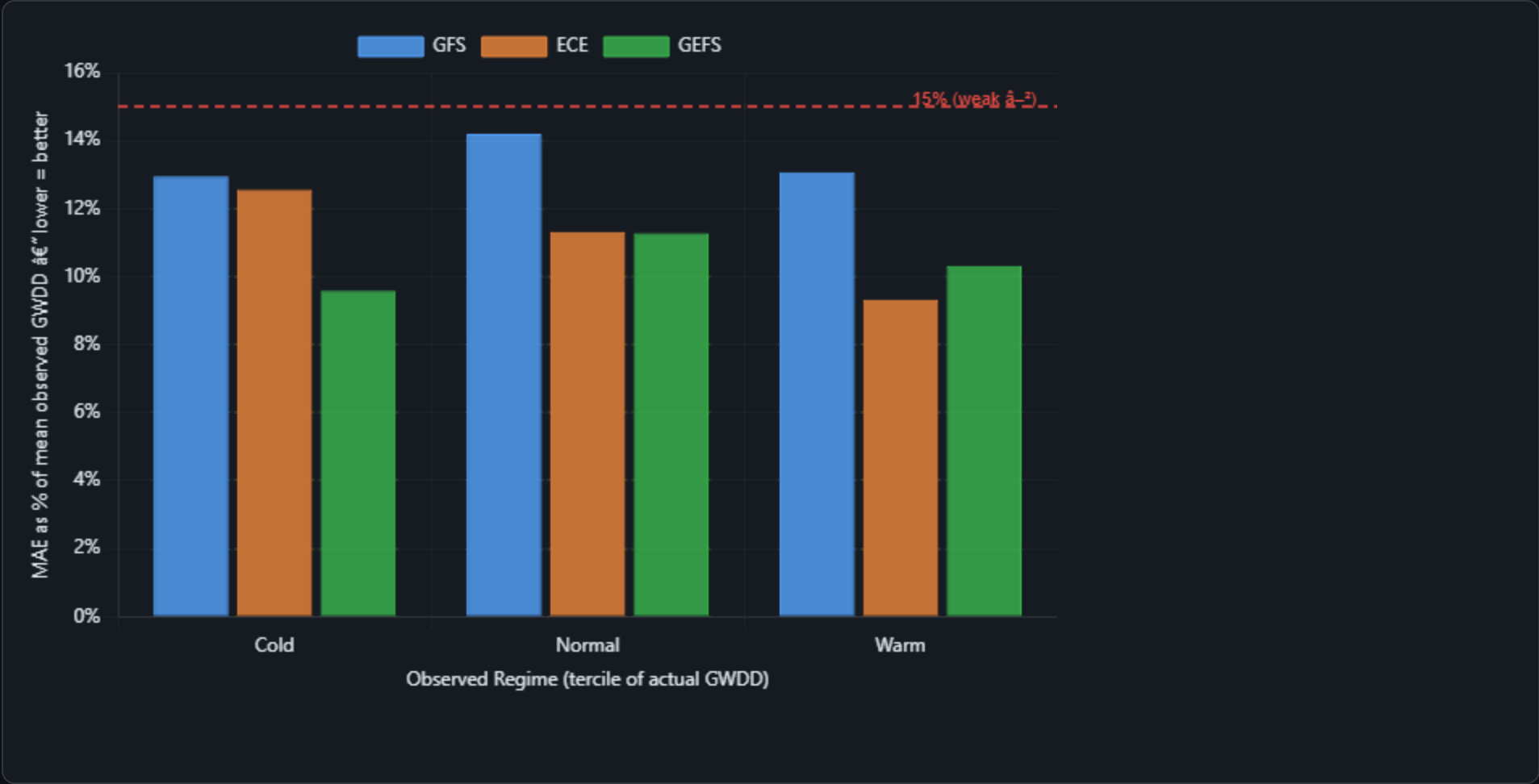
Model Divergence (GFS vs ECE)

Rolling 30-day average of $|GFS - ECE|$ spread per ForecastDate, by window. Wider spread = more model disagreement.



Regime Performance

Lower bars = better forecasts. MAE expressed as a **percentage of the mean observed GWDD in that regime**, so cold/normal/warm groups are directly comparable despite their very different absolute magnitudes. Regimes are terciles of observed GWDD: Cold ≈ 9.12 (summer baseline), Normal 9.12–15.84 (shoulder), Warm > 15.84 (winter peak demand). The dashed line marks 15% ≈ a rough "weak forecast" threshold; bars above it are unreliable.



Lead-Day Sanity Check

All forecast stats broken out by integer lead day, **including lead 0 (nowcast – same-day "diagnosis") and lead 15 (longest horizon)**. Lead 0 should be near-perfect (model's own initialization); lead 15 should be the worst. Anything that breaks that pattern is a data issue worth investigating.

Model	Lead	N	MAE	RMSE	Bias	Corr	±1 Hit%	±2 Hit%
GFS	0 (nowcast)	1066	0.991	1.197	-0.813	0.995	55.2	91.7
GFS	1	1066	1.067	1.298	-0.814	0.994	53.4	87.0
GFS	2	1066	1.145	1.422	-0.818	0.992	52.9	83.9
GFS	3	1066	1.196	1.517	-0.791	0.990	51.5	81.5
GFS	4	1066	1.216	1.589	-0.697	0.987	52.3	80.1
GFS	5	1066	1.294	1.741	-0.596	0.982	51.9	77.4
GFS	6	1066	1.490	2.090	-0.511	0.971	47.4	74.5
GFS	7	1066	1.661	2.368	-0.538	0.962	42.3	71.4
GFS	8	1065	1.964	2.793	-0.569	0.946	37.7	65.0
GFS	9	1065	2.199	3.173	-0.609	0.929	36.2	61.6
GFS	10	1065	2.454	3.477	-0.601	0.913	32.9	57.1
GFS	11	1064	2.665	3.898	-0.560	0.888	31.7	55.0
GFS	12	1063	2.745	4.125	-0.546	0.874	33.1	56.3
GFS	13	1063	2.902	4.316	-0.450	0.860	30.2	53.4
GFS	14	1060	3.054	4.581	-0.322	0.841	30.1	52.7
GFS	15 (longest)	541	2.691	3.929	-0.126	0.831	30.7	55.8
ECE	0 (nowcast)	1034	0.900	1.067	-0.672	0.997	59.0	97.3
ECE	1	1033	0.903	1.059	-0.539	0.997	58.7	97.4
ECE	2	1032	0.935	1.103	-0.466	0.996	56.0	96.7
ECE	3	1032	0.949	1.148	-0.376	0.995	57.5	94.6
ECE	4	1031	1.003	1.233	-0.333	0.994	56.6	91.4
ECE	5	1030	1.130	1.404	-0.325	0.990	50.9	87.2
ECE	6	1028	1.251	1.614	-0.264	0.985	48.2	82.6
ECE	7	1027	1.377	1.873	-0.279	0.978	46.7	78.5

Model	Lead	N	MAE	RMSE	Bias	Corr	Â±1 Hit%	Â±2 Hit%
ECE	8	1026	1.534	2.088	-0.266	0.972	44.3	76.1
ECE	9	1025	1.692	2.352	-0.260	0.963	41.1	71.6
ECE	10	1025	1.870	2.650	-0.231	0.952	38.9	67.1
ECE	11	1023	1.992	2.887	-0.188	0.943	39.5	67.3
ECE	12	1022	2.159	3.105	-0.148	0.933	36.7	63.2
ECE	13	1018	2.325	3.345	-0.118	0.921	36.1	61.4
ECE	14	495	1.990	2.821	-0.211	0.920	37.0	66.9
GEFS	0 (nowcast)	1023	0.717	0.869	-0.629	0.998	72.2	99.4
GEFS	1	1022	0.768	0.966	-0.624	0.996	70.9	96.7
GEFS	2	1021	0.825	1.038	-0.609	0.995	67.6	93.8
GEFS	3	1020	0.874	1.126	-0.564	0.993	64.9	91.8
GEFS	4	1020	0.941	1.237	-0.502	0.991	62.6	90.1
GEFS	5	1019	1.054	1.457	-0.432	0.986	59.6	86.1
GEFS	6	1013	1.201	1.705	-0.361	0.981	57.5	82.6
GEFS	7	1008	1.363	1.975	-0.369	0.974	52.5	77.8
GEFS	8	1001	1.550	2.272	-0.385	0.965	49.5	74.6
GEFS	9	998	1.754	2.589	-0.405	0.954	44.4	72.6
GEFS	10	996	1.938	2.848	-0.365	0.944	41.0	68.6
GEFS	11	992	2.061	3.037	-0.325	0.936	39.0	65.7
GEFS	12	990	2.158	3.215	-0.277	0.928	39.6	63.8
GEFS	13	986	2.286	3.380	-0.210	0.920	39.0	60.5
GEFS	14	976	2.381	3.514	-0.130	0.912	35.5	61.7
GEFS	15 (longest)	489	1.995	2.984	-0.118	0.907	43.4	69.1

Bias by GWDD Magnitude

Mean Bias (forecast $\hat{y}$ - actual) and MAE binned by the **magnitude of the observed actual GWDD**. Tells you whether models systematically under- or over-shoot when actuals are small vs. large. A 15-GWDD miss when the actual was 35 (winter peak) is a different story than a 15-GWDD miss when the actual was 10 (shoulder).

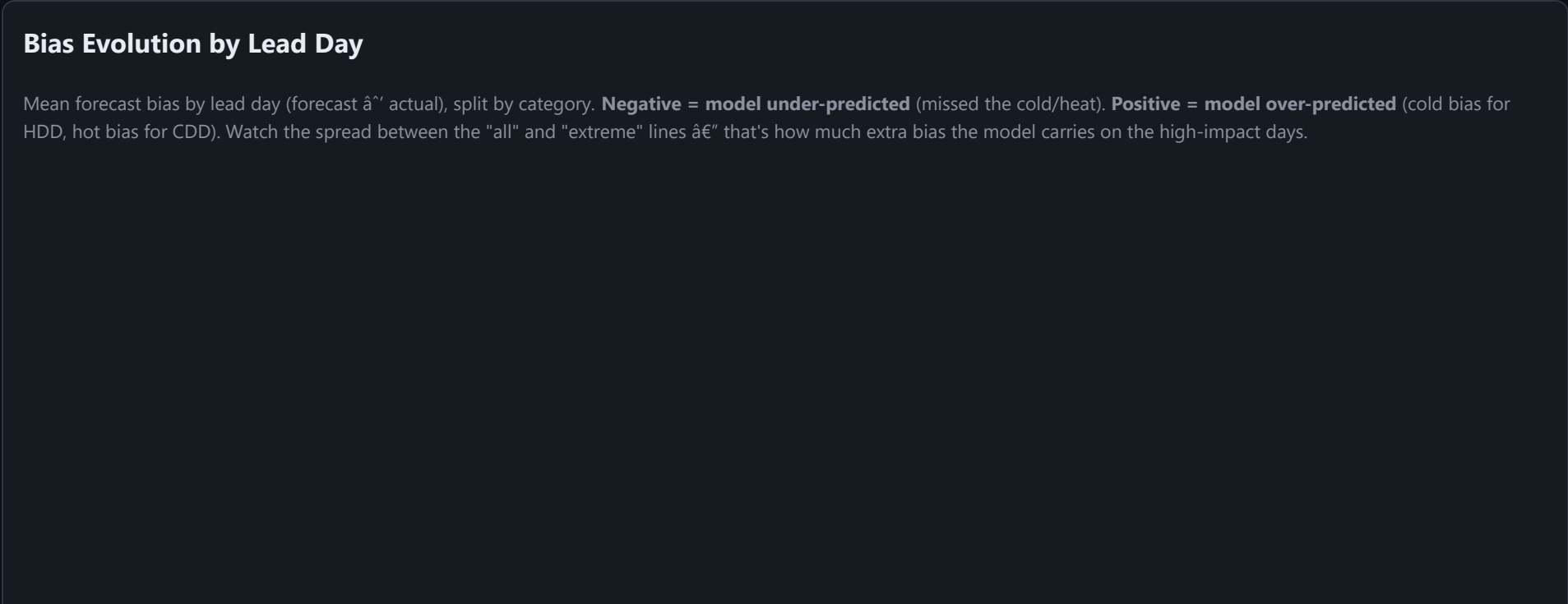


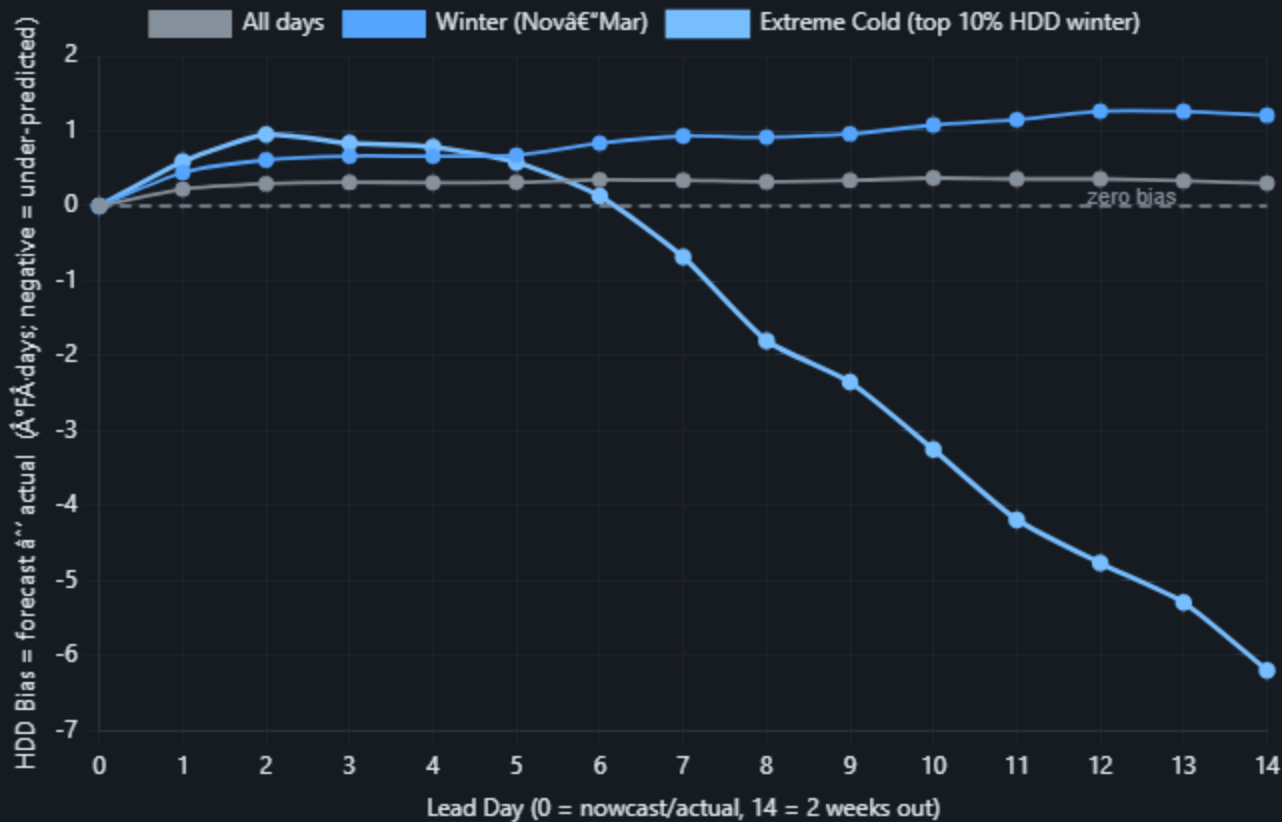
Model	Window	Bin	N	MeanActual	Bias	MAE	MAE %
GFS	1-5d	25-30	380	27.54	-1.983	2.091	7.59%
GFS	1-5d	30+	355	35.85	-1.490	1.762	4.91%
ECE	1-5d	5-10	2112	8.23	-0.987	1.038	12.62%
ECE	1-5d	10-15	1153	11.71	-0.545	0.842	7.19%
ECE	1-5d	15-20	670	17.46	+0.142	0.900	5.16%
ECE	1-5d	20-25	488	22.24	+0.367	0.979	4.40%
ECE	1-5d	25-30	380	27.54	+0.177	0.919	3.34%
ECE	1-5d	30+	355	35.85	+0.752	1.356	3.78%
GEFS	1-5d	5-10	2099	8.23	-0.255	0.568	6.90%
GEFS	1-5d	10-15	1134	11.69	-0.460	0.820	7.01%
GEFS	1-5d	15-20	654	17.46	-0.923	1.235	7.07%
GEFS	1-5d	20-25	482	22.24	-0.991	1.326	5.96%
GEFS	1-5d	25-30	378	27.54	-1.045	1.372	4.98%
GEFS	1-5d	30+	355	35.85	-0.713	1.311	3.66%

KDCA – ECE Forecast Bias on Extreme Events

Single-station ECE 51-member ensemble (00Z, 6-hour steps, 2020–2025) for Reagan National (KDCA). Temperatures converted $^{\circ}\text{C} \rightarrow ^{\circ}\text{F}$, then $\text{HDD} = \max(0, 65 - T_f)$ and $\text{CDD} = \max(0, T_f - 65)$. Lead 0 (the same-day "nowcast") serves as the observed proxy. **Bias = forecast_L – actual₀**; **NEGATIVE bias = model UNDER-predicted** (forecast was warmer than reality – missed the cold for HDD, or cooler than reality – missed the heat for CDD). **POSITIVE bias = model OVER-predicted** (cold bias for HDD, hot bias for CDD). The interesting question: does the bias get worse on the days that actually matter – the coldest winter and hottest summer days?

Date range	2020-01-01 to 2025-12-31
Total valid dates	2190
Winter days	907
Summer days	1283
Extreme cold winter days (top 10% HDD)	91
Extreme hot summer days (top 10% CDD)	129
Winter top-10% HDD threshold (F-days)	34.69
Summer top-10% CDD threshold (F-days)	14.48





Top Extreme Events – Forecast Evolution

Top 25 events per category, showing how the forecast for that day looked at leads 1, 3, 5, 7, 10, 14 vs. the lead-0 realization. Bias columns = forecast - actual (negative = model under-predicted the extreme).

ValidDate	Month	HDD ₀	L1	L3	L5	L7	L10	L14	Bias L1	Bias L3	Bias L5	Bias L7	Bias L10	Bias L14
2022-12-24	Dec	53.56	50.87	49.97	42.13	47.31	40.52	34.48	-2.70	-3.60	-11.43	-6.25	-13.05	-19.08
2025-01-21	Jan	51.72	54.39	54.78	55.35	49.82	36.06	34.03	+2.67	+3.06	+3.63	-1.90	-15.66	-17.69
2025-01-22	Jan	51.51	53.15	53.84	54.99	53.92	49.24	44.30	+1.64	+2.32	+3.48	+2.40	-2.27	-7.21
2025-01-23	Jan	48.66	50.78	52.93	56.67	51.15	46.73	41.04	+2.12	+4.27	+8.01	+2.49	-1.93	-7.62
2025-12-15	Dec	46.70	46.19	42.32	42.54	38.93	33.89	28.92	-0.50	-4.37	-4.16	-7.77	-12.81	-17.77

ValidDate	Month	HDD ₀	L1	L3	L5	L7	L10	L14	Bias L1	Bias L3	Bias L5	Bias L7	Bias L10	Bias L14
2024-01-17	Jan	45.57	45.23	42.33	42.68	43.31	36.96	32.32	-0.34	-3.24	-2.88	-2.25	-8.61	-13.25
2022-12-25	Dec	44.84	43.48	44.88	44.79	42.59	43.35	34.26	-1.35	+0.04	-0.04	-2.25	-1.49	-10.57
2022-01-30	Jan	44.70	45.15	45.69	46.68	40.97	34.76	36.33	+0.45	+0.99	+1.98	-3.73	-9.95	-8.37
2025-01-16	Jan	44.34	45.13	45.84	46.79	47.01	44.37	42.39	+0.78	+1.50	+2.45	+2.66	+0.02	-1.95
2025-02-20	Feb	44.24	40.84	41.62	41.73	41.07	31.99	32.34	-3.40	-2.62	-2.52	-3.17	-12.25	-11.90
2023-02-04	Feb	44.12	45.51	46.01	44.56	39.03	39.57	29.98	+1.40	+1.89	+0.45	-5.08	-4.54	-14.13
2024-01-21	Jan	44.06	44.86	43.70	44.05	45.48	42.93	36.29	+0.80	-0.36	-0.01	+1.42	-1.13	-7.77
2025-01-09	Jan	43.85	45.06	46.41	46.99	46.37	40.56	34.96	+1.21	+2.56	+3.13	+2.52	-3.29	-8.90
2022-01-22	Jan	42.99	42.85	44.20	45.10	43.84	40.24	34.48	-0.14	+1.22	+2.12	+0.85	-2.75	-8.51
2022-01-16	Jan	42.90	41.51	43.08	43.94	37.09	35.61	33.45	-1.40	+0.18	+1.03	-5.80	-7.29	-9.45
2024-01-20	Jan	42.80	42.05	42.69	41.69	41.54	37.81	35.91	-0.75	-0.11	-1.11	-1.26	-4.98	-6.88
2025-01-20	Jan	42.72	44.28	45.44	43.76	38.67	33.42	37.10	+1.56	+2.72	+1.05	-4.04	-9.29	-5.62
2022-01-08	Jan	42.49	41.73	41.87	40.92	39.12	31.25	26.88	-0.77	-0.63	-1.57	-3.38	-11.25	-15.62
2025-02-19	Feb	42.40	43.02	43.41	40.17	39.96	33.95	27.96	+0.63	+1.01	-2.23	-2.44	-8.44	-14.44
2022-01-21	Jan	42.36	43.76	43.22	44.84	41.15	33.95	32.05	+1.40	+0.85	+2.48	-1.22	-8.41	-10.30
2025-01-10	Jan	41.97	44.44	46.19	45.69	45.02	42.47	42.23	+2.48	+4.23	+3.73	+3.05	+0.50	+0.26
2025-01-15	Jan	41.72	43.41	46.06	48.00	45.46	40.67	39.21	+1.69	+4.34	+6.28	+3.74	-1.05	-2.51
2024-01-22	Jan	41.58	42.58	42.99	42.06	40.03	38.97	32.83	+1.00	+1.41	+0.49	-1.54	-2.61	-8.74
2021-02-21	Feb	40.97	41.77	43.58	41.19	34.98	35.61	33.59	+0.81	+2.61	+0.23	-5.99	-5.36	-7.38
2025-01-07	Jan	40.94	40.84	40.82	40.81	36.37	37.56	31.90	-0.10	-0.12	-0.13	-4.57	-3.38	-9.04

Extreme Events

Granular row-level lookups are split into three sections: (1) most extreme forecasts (standardized anomaly, z-scored by month), (2) most extreme observed actuals (top & bottom of the history), (3) biggest forecast misses per individual lead day 10/15.

Most Extreme Forecasts (top 50 by |z|)

Anomaly standardized by month-of-year std-dev so summer/winter are comparable. Higher |z| = more unusual call.

Model	RunDate	ForecastDate	Lead	Forecast	Anomaly	z-score	Actual	Error
GFS	2023-08-14	2023-08-24	10	17.57	+8.60	+5.48	12.69	+4.88
GFS	2023-06-24	2023-07-09	15	16.78	+7.57	+4.87	7.85	+8.94
GFS	2023-08-14	2023-08-25	11	16.34	+7.31	+4.66	11.94	+4.40
GFS	2023-08-14	2023-08-23	9	16.07	+7.12	+4.54	11.88	+4.18
GEFS	2023-10-29	2023-10-31	2	21.63	+7.87	+4.51	21.82	-0.19
GFS	2023-08-28	2023-09-08	11	14.50	+6.86	+4.49	10.29	+4.21
GFS	2023-08-30	2023-09-11	12	14.38	+6.86	+4.49	8.41	+5.97
GEFS	2023-10-30	2023-10-31	1	21.51	+7.74	+4.44	21.82	-0.31
GEFS	2023-10-31	2023-10-31	0	21.49	+7.72	+4.43	21.82	-0.33
GFS	2023-07-29	2023-08-13	15	16.18	+6.93	+4.42	10.78	+5.41
GFS	2023-07-24	2023-08-08	15	16.51	+6.90	+4.40	9.45	+7.07
GFS	2024-10-18	2024-10-31	13	6.13	-7.65	-4.39	8.80	-2.66
GFS	2023-08-19	2023-08-24	5	15.81	+6.83	+4.36	12.69	+3.12
GFS	2023-08-30	2023-09-10	11	14.27	+6.64	+4.35	8.12	+6.16
GEFS	2023-10-28	2023-10-31	3	21.33	+7.56	+4.34	21.82	-0.49
GFS	2023-08-29	2023-09-06	8	14.66	+6.61	+4.33	10.60	+4.06
GFS	2023-08-26	2023-09-04	9	14.85	+6.61	+4.32	10.57	+4.28
GFS	2023-08-29	2023-09-05	7	14.83	+6.60	+4.32	11.17	+3.66
GFS	2023-10-25	2023-10-31	6	21.28	+7.51	+4.31	21.82	-0.54
GFS	2024-10-21	2024-10-31	10	6.32	-7.47	-4.29	8.80	-2.48
GFS	2025-10-14	2025-10-28	14	5.99	-7.45	-4.27	12.47	-6.48

Model	RunDate	ForecastDate	Lead	Forecast	Anomaly	z-score	Actual	Error
ECE	2023-10-30	2023-10-31	1	21.13	+7.36	+4.22	21.82	-0.69
ECE	2023-10-28	2023-10-31	3	21.08	+7.31	+4.19	21.82	-0.74
GFS	2023-08-28	2023-09-07	10	14.16	+6.40	+4.19	10.55	+3.61
GFS	2024-10-20	2024-10-30	10	6.19	-7.26	-4.17	7.94	-1.75
ECE	2023-10-29	2023-10-31	2	21.00	+7.24	+4.15	21.82	-0.82
GFS	2023-06-26	2023-07-11	15	15.55	+6.44	+4.15	10.18	+5.37
GFS	2023-07-19	2023-08-02	14	16.40	+6.51	+4.15	10.60	+5.80
GEFS	2023-10-27	2023-10-31	4	21.00	+7.23	+4.15	21.82	-0.82
ECE	2023-10-31	2023-10-31	0	20.98	+7.22	+4.14	21.82	-0.84
GFS	2024-08-13	2024-08-28	15	14.90	+6.47	+4.13	10.68	+4.22
GFS	2023-08-20	2023-08-24	4	15.43	+6.46	+4.12	12.69	+2.74
GFS	2023-10-15	2023-10-29	14	6.16	-7.16	-4.11	14.61	-8.44
GFS	2024-10-19	2024-10-28	9	6.29	-7.14	-4.10	10.86	-4.56
GFS	2024-10-21	2024-10-30	9	6.33	-7.12	-4.09	7.94	-1.61
ECE	2023-10-25	2023-10-31	6	20.86	+7.10	+4.07	21.82	-0.96
GFS	2023-08-27	2023-09-04	8	14.46	+6.21	+4.07	10.57	+3.88
GEFS	2023-10-26	2023-10-31	5	20.83	+7.06	+4.05	21.82	-0.99
GEFS	2023-10-25	2023-10-31	6	20.81	+7.04	+4.04	21.82	-1.01
GFS	2024-10-22	2024-10-31	9	6.77	-7.02	-4.03	8.80	-2.03
GFS	2023-06-15	2023-06-28	13	13.23	+4.74	+4.00	9.51	+3.72
ECE	2023-10-27	2023-10-31	4	20.74	+6.97	+4.00	21.82	-1.08
GFS	2023-08-18	2023-08-24	6	15.24	+6.26	+3.99	12.69	+2.55
GFS	2023-08-26	2023-09-10	15	13.68	+6.06	+3.97	8.12	+5.56
GFS	2024-10-20	2024-10-31	11	6.88	-6.91	-3.96	8.80	-1.92
GFS	2024-10-22	2024-10-30	8	6.58	-6.88	-3.95	7.94	-1.36
GFS	2023-08-28	2023-09-05	8	14.24	+6.01	+3.94	11.17	+3.07
GFS	2024-07-21	2024-08-04	14	15.63	+6.16	+3.93	9.96	+5.67
GFS	2023-07-21	2023-07-30	9	15.65	+6.07	+3.91	10.65	+5.00

Model	RunDate	ForecastDate	Lead	Forecast	Anomaly	z-score	Actual	Error
GFS	2023-07-25	2023-07-28	3	15.44	+6.06	+3.90	13.09	+2.35

Most Extreme Observed Actuals

Top 15 highest and 15 lowest GWDD days in the history file.

Side	Date	Actual GWDD
Lowest	2023-06-01	3.833
Lowest	2024-09-27	5.561
Lowest	2024-09-26	5.631
Lowest	2025-09-27	5.748
Lowest	2025-09-25	5.758
Lowest	2025-09-24	5.815
Lowest	2025-09-28	5.881
Lowest	2025-09-29	5.887
Lowest	2025-09-30	5.900
Lowest	2023-10-05	5.923
Lowest	2024-09-28	6.005
Lowest	2024-05-17	6.011
Lowest	2024-09-25	6.120
Lowest	2023-06-16	6.166
Lowest	2024-09-29	6.188
Highest	2025-01-21	47.477
Highest	2025-01-20	44.947
Highest	2024-01-16	44.723
Highest	2024-01-15	44.536
Highest	2026-01-26	43.266
Highest	2025-01-22	43.131
Highest	2026-01-25	43.049

Side	Date	Actual GWDD
Highest	2026-01-24	42.811
Highest	2024-01-20	41.966
Highest	2026-01-31	41.785
Highest	2024-01-17	41.358
Highest	2026-01-27	41.223
Highest	2024-01-14	41.089
Highest	2025-02-20	40.846
Highest	2025-02-19	40.650

Biggest Misses €” Per Lead Day 10€”15

Top 10 largest |Error| rows for the selected lead day. Drill into specific events.

Model	RunDate	ForecastDate	Forecast	Actual	Error	Error	Anomaly
GFS	2024-01-11	2024-01-26	37.33	18.66	+18.67	18.67	+8.08
GFS	2024-11-16	2024-12-01	10.53	28.71	-18.18	18.18	-11.48
GFS	2024-01-02	2024-01-17	23.82	41.36	-17.54	17.54	-6.45
GFS	2024-01-01	2024-01-16	27.26	44.72	-17.47	17.47	-2.99
GFS	2023-10-17	2023-11-01	7.77	22.90	-15.13	15.13	-6.00
GEFS	2024-01-01	2024-01-16	30.48	44.72	-14.25	14.25	+0.23
GFS	2024-01-05	2024-01-20	27.77	41.97	-14.20	14.20	-2.46
GFS	2024-01-10	2024-01-25	31.95	18.20	+13.75	13.75	+2.24
GFS	2024-01-12	2024-01-27	33.88	20.52	+13.36	13.36	+5.14
GEFS	2023-12-31	2024-01-15	31.47	44.54	-13.06	13.06	+1.37
GFS	2023-11-12	2023-11-27	14.71	27.05	-12.34	12.34	-6.46
GFS	2024-02-14	2024-02-29	13.55	25.64	-12.09	12.09	-11.27
GEFS	2023-12-30	2024-01-14	29.34	41.09	-11.75	11.75	-0.22
GEFS	2024-01-02	2024-01-17	30.11	41.36	-11.24	11.24	-0.15
GEFS	2024-11-15	2024-11-30	19.18	29.92	-10.74	10.74	-1.98

Model	RunDate	ForecastDate	Forecast	Actual	Error	Error	Anomaly
GEFS	2024-02-14	2024-02-29	15.10	25.64	-10.54	10.54	-9.72
GEFS	2023-10-17	2023-11-01	12.80	22.90	-10.10	10.10	-0.97
GEFS	2023-10-16	2023-10-31	11.94	21.82	-9.88	9.88	-1.37
GEFS	2024-01-05	2024-01-20	32.17	41.97	-9.79	9.79	+1.95
GEFS	2024-01-11	2024-01-26	28.05	18.66	+9.39	9.39	-1.20

Market Insights

- At the 1-5d horizon, **GEFS** is the most accurate (RMSE 1.18, MAE 0.89), while GFS trails (RMSE 1.52).
- At the 6-10d horizon, **ECE** is the most accurate (RMSE 2.15, MAE 1.54), while GFS trails (RMSE 2.83).
- At the 11-15d horizon, **ECE** is the most accurate (RMSE 3.08, MAE 2.13), while GFS trails (RMSE 4.20).
- The largest systematic bias is **GFS** at 1-5d: -0.74 GWDD on average — runs low (forecasts warmer than reality).
- Seasonal bias warnings ($|\text{bias}| > 1$ GWDD): GFS/1-5d/SON (-1.00), GFS/11-15d/DJF (-1.75), GFS/6-10d/DJF (-1.43), GFS/1-5d/DJF (-1.56), ECE/1-5d/JJA (-1.02).
- **GFS** is the most volatile model run-to-run (mean $|\text{Chg1}| = 1.28$); **ECE** is the most stable (0.17).
- Run-to-run drift (mean signed Chg1): GFS trends lower (-0.048); ECE no consistent drift (-0.010); GEFS no consistent drift (-0.014).
- RMSE degradation from 1-5d to 11-15d: .
- The hardest-to-predict combo is **GFS** at 11-15d during **Warm** regimes (MAE 4.94).
- On extreme-temperature days (top/bottom 10% of actuals), **GEFS** at 1-5d performs best (MAE 0.92); GFS at 11-15d performs worst (3.97).